

Final Capstone Project

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OUTLINE



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EXECUTIVE SUMMARY

- Companies are making space travel affordable for everyone. Virgin Galactic is providing suborbital spaceflights. Rocket Lab is a small satellite provider. Blue Origin manufactures sub-orbital and orbital reusable rockets. Perhaps the most successful is SpaceX.
- One reason SpaceX can do this is the rocket launches are relatively inexpensive. SpaceX advertises Falcon 9 rocket launches on its website with a cost of 62 million dollars; other providers cost upwards of 165 million dollars each, much of the savings is because SpaceX can reuse the first stage.
- Space Y would like to compete with SpaceX founded by Billionaire industrialist Allon Musk.
- Therefore, if we can determine if the first stage will land, we can determine the cost of a launch.
- Determine the price of each launch by:
 - Gathering information about Space X and creating dashboards.
 - Determine if SpaceX will reuse the first stage.
 - To determine if the first stage will land successfully, train a machine learning model and use public information to predict if SpaceX will reuse the first stage.



INTRODUCTION

- This analysis will determine if the first stage will land successfully, so the cost of a launch can be determined
- This information can be used if an alternate company wants to bid against SpaceX for a rocket launch
- Collect data on the Falcon 9 first-stage landings.
- Predict if the Falcon 9 first stage will land successfully
- Determine the price of each launch



METHODOLOGY

- Collect data on the Falcon 9 first-stage landings.
- Use a RESTful API and web scraping - This API will give us data about launches, including information about the rocket used, payload delivered, launch specifications, landing specifications, and landing outcome.
- Convert the data into a dataframe and then perform some data wrangling.
- Create scatter plots and bar charts by writing Python code to analyze data in a Pandas data frame
- Write Python code to conduct exploratory data analysis by manipulating data in a Pandas data frame
- Create and execute SQL queries to select and sort data
- Use data visualization skills to visualize the data and extract meaningful patterns to guide the modeling process



Predictive Analysis And Data Wrangling Methodology

- Collecting and wrangling data
- Collect and make sure the data is in the correct format from an API
 - Make a get request to the SpaceX API and basic data wrangling and formatting
 - Request to the SpaceX API
 - Clean the requested data
 - Decode the response content as a .json and turn it into a Pandas dataframe using `.json_normalize()`
- Use API to get information about the launches using the IDs given for each launch – using columns data stored in a lists and used to create a new dataframe
 - rocket – learn the booster name
 - payloads – learn the mass of the payload and the orbit that it is going to
 - launchpad – learn the name of the launch site being used, the longitude, and the latitude
 - cores – learn the outcome of the landing, the type of the landing, number of flights with that core, whether gridfins were used, whether the core is reused, whether the legs were used, the landing pad used, the block of the which is a number used to separate version of cores,, the number of times this specific core has been reused,, and the serial of the core
- Then create a dataset using the data that was obtained and combine the columns into a dictionary



Predictive Analysis Methodology

A new dataframe is created that will only have the Falcon 9 launches

```
#Global variables
BoosterVersion = []
PayloadMass = []
Orbit = []
LaunchSite = []
Outcome = []
Flights = []
GridFins = []
Reused = []
Legs = []
LandingPad = []
Block = []
ReusedCount = []
Serial = []
Longitude = []
Latitude = []
```

```
launch_dict = {'FlightNumber': list(data['flight_number']),
               'Date': list(data['date']),
               'BoosterVersion':BoosterVersion,
               'PayloadMass':PayloadMass,
               'Orbit':Orbit,
               'LaunchSite':LaunchSite,
               'Outcome':Outcome,
               'Flights':Flights,
               'GridFins':GridFins,
               'Reused':Reused,
               'Legs':Legs,
               'LandingPad':LandingPad,
               'Block':Block,
               'ReusedCount':ReusedCount,
               'Serial':Serial,
               'Longitude': Longitude,
               'Latitude': Latitude}
```

Data Collection and Data Wrangling - Methodology

- Table before missing values are removed

FlightNumber	Date	BoosterVersion	PayloadMass	Orbit	LaunchSite	Outcome	Fligh
4	1	2010-06-04	Falcon 9	NaN	LEO	CCSFS SLC 40	None None
5	2	2012-05-22	Falcon 9	525.0	LEO	CCSFS SLC 40	None None
6	3	2013-03-01	Falcon 9	677.0	ISS	CCSFS SLC 40	None None
7	4	2013-09-29	Falcon 9	500.0	PO	VAFB SLC 4E	False Ocean
8	5	2013-12-03	Falcon 9	3170.0	GTO	CCSFS SLC 40	None None
...	...	...	...	...	...	...	...
89	86	2020-09-03	Falcon 9	15600.0	VLEO	KSC LC 39A	True ASDS
90	87	2020-10-06	Falcon 9	15600.0	VLEO	KSC LC 39A	True ASDS
91	88	2020-10-18	Falcon 9	15600.0	VLEO	KSC LC 39A	True ASDS
92	89	2020-10-24	Falcon 9	15600.0	VLEO	CCSFS SLC 40	True ASDS

- Remove the missing values in the dataframe

The Landing Pad column will retain None values to represent when landing pads were not used.

```
FlightNumber    0
Date            0
BoosterVersion  0
PayloadMass     5
Orbit           0
LaunchSite      0
Outcome         0
Flights         0
GridFins        0
Reused          0
Legs            0
LandingPad      26
Block           0
ReusedCount     0
Serial          0
Longitude       0
Latitude        0
dtype: int64
```



Data Collection and Data Wrangling - Methodology

PayloadMass changed to zero

There are no missing values in the dataset except for Landing Pad

```
FlightNumber      0
Date              0
BoosterVersion    0
PayloadMass       0
Orbit             0
LaunchSite        0
Outcome           0
Flights           0
GridFins          0
Reused            0
Legs              0
LandingPad        26
Block             0
ReusedCount       0
Serial            0
Longitude         0
Latitude          0
dtype: int64
```



RESULTS

- The SpaceX launch data will be used and is gathered from an API, specifically the SpaceX REST API. This API will give us data about launches, including information about the rocket used, payload delivered, launch specifications, landing specifications, and landing outcome. Our goal is to use this data to predict whether SpaceX will attempt to land a rocket or not.
- This will include Preprocessing, allowing us to standardize our data, allowing us to split our data into training and testing data. We will train the model and perform Grid Search, allowing us to find the hyperparameters that allow a given algorithm to perform best. Using the best hyperparameter values, we will determine the model with the best accuracy using the training data. Will test Logistic Regression, Support Vector machines, Decision Tree Classifier, and K-nearest neighbors and output the confusion matrix.
- Another data source for obtaining Falcon 9 Launch data is web scraping related Wiki pages. Python BeautifulSoup package is used to web scrape HTML tables that contain valuable Falcon 9 launch records. Then the data from those tables will be parsed and convert into a Pandas data frame for further visualization and analysis. The data will be from raw data into a clean dataset which provides meaningful data on the situation we are trying to address: Wrangling Data using an API, Sampling Data, and Dealing with Nulls.
- We have the different end points, for example: /capsules and /cores that be used and will be working with the endpoint `api.spacexdata.com/v4/launches/past`. The SpaceX REST API endpoints, or URL, starts with `api.spacedata.com/v4/`.
- Use a RESTful API and web scraping
- Convert the data into a dataframe and then perform some data wrangling.
- Create scatter plots and bar charts by writing Python code to analyze data in a Pandas data frame

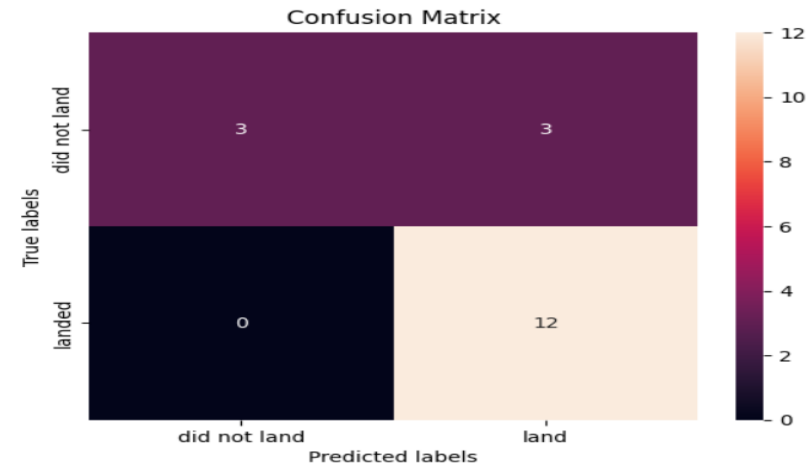


Predicate Analysis Classification

- Perform exploratory Data Analysis and determine Training Labels
- create a column for the class
- Standardize the data
- Split into training data and test data
- -Find best Hyperparameter for SVM, Classification Trees and Logistic Regression
- Find the method performs best using test data
 - The best performing method is Decision Tree Classifier
- We split the data into training and testing data using the function `train_test_split`. The training data is divided into validation data, a second set used for training data; then the models are trained and hyperparameters are selected using the function `GridSearchCV`.
- # Accuracy scores of different methods
`accuracy_lr = logreg_cv.score(X_test, Y_test)`
`accuracy_svm = svm_cv.score(X_test, Y_test)`
`accuracy_tree = tree_cv.score(X_test, Y_test)`
`accuracy_knn = knn_cv.score(X_test, Y_test)`
Find the method with the highest accuracy
`best_accuracy = max(accuracy_lr, accuracy_svm, accuracy_tree, accuracy_knn)`
- # Determine the best performing method if
`best_accuracy == accuracy_lr: best_method = "Logistic Regression"`
`elif best_accuracy == accuracy_svm: best_method = "Support Vector Machine (SVM)"`
`elif best_accuracy == accuracy_tree: best_method = "Decision Tree Classifier"`
`else: best_method = "K Nearest Neighbors (KNN)"`

Predicate Analysis Classification

- Calculate the accuracy on the test data using the method score
- Accuracy on test data: 0.8333333333333334
- Create a logistic regression object then create a GridSearchCV object logreg_cv with cv = 10. Fit the object to find the best parameters from the dictionary parameters.
- The output GridSearchCV object for logistic regression. Display the best parameters using the data attribute best_params_ and the accuracy on the validation data using the data attribute best_score_.
- Tuned hyperparameters : (best parameters)
{'C': 1.0, 'gamma': 0.03162277660168379, 'kernel': 'sigmoid'}
- Accuracy : 0.8482142857142856



- Examining the confusion matrix, we see that logistic regression can distinguish between the different classes. We see that the problem is false positives.
- Overview:
- True Positive - 12 (True label is landed, Predicted label is also landed)
- False Positive - 3 (True label is not landed, Predicted label is landed)

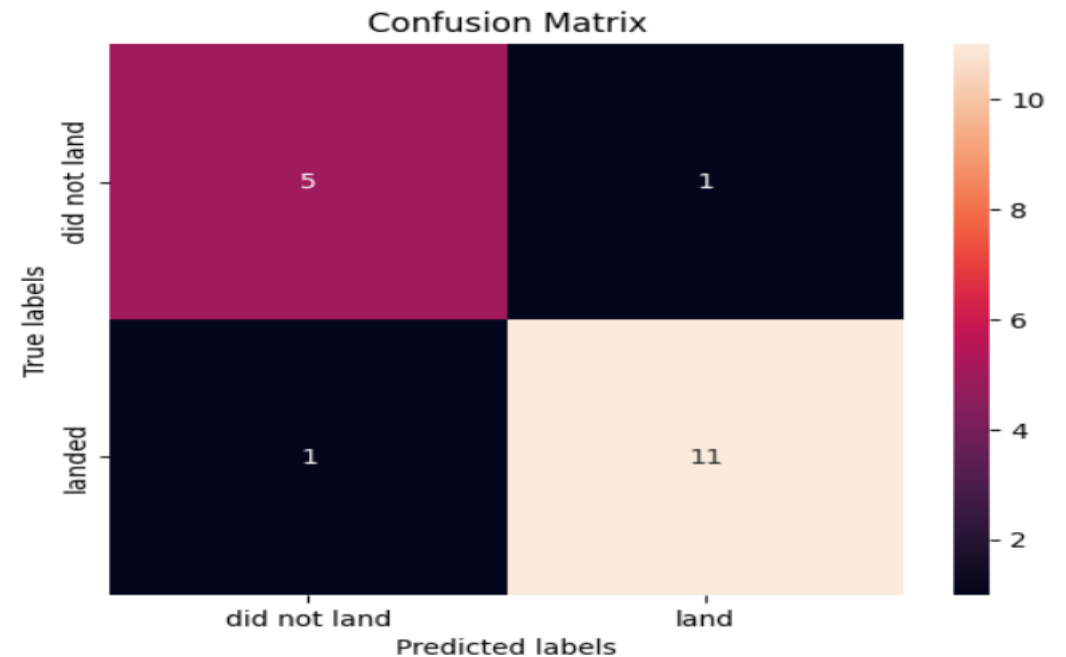
Predicate Analysis Classification

Create a support vector machine object then create a GridSearchCV object svm_cv with cv = 10. Fit the object to find the best parameters from the dictionary parameters.

Tuned hyperparameters : (best parameters) {'C': 1.0, 'gamma': 0.03162277660168379, 'kernel': 'sigmoid'}

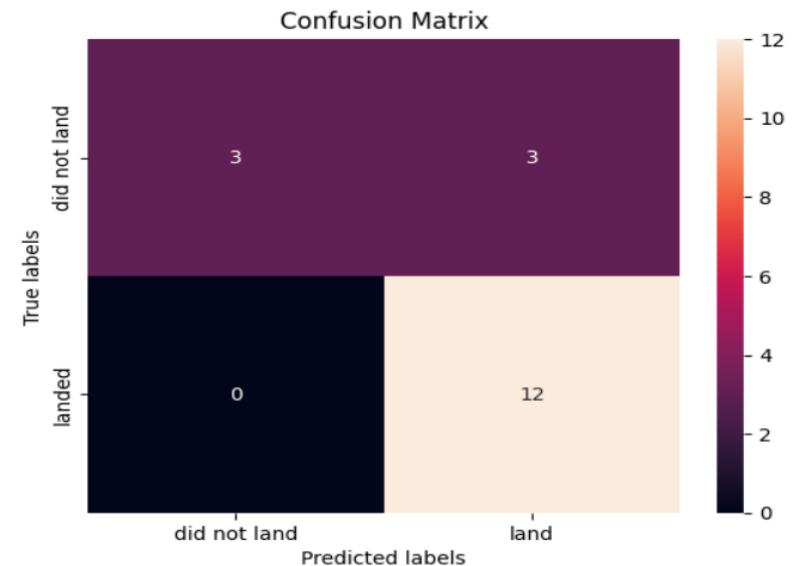
accuracy : 0.8482142857142856

- Calculate the accuracy of tree_cv on the test data using the method score:
- Accuracy on test data: 0.8888888888888888



Predicate Analysis Classification

- Create a k nearest neighbors object then create a GridSearchCV object knn_cv with cv = 10.
Knn accuracy = knn_cv.score(X_test, Y_test)
print("Accuracy on test data:", accuracy) -
Accuracy on test data: 0.8333333333333334

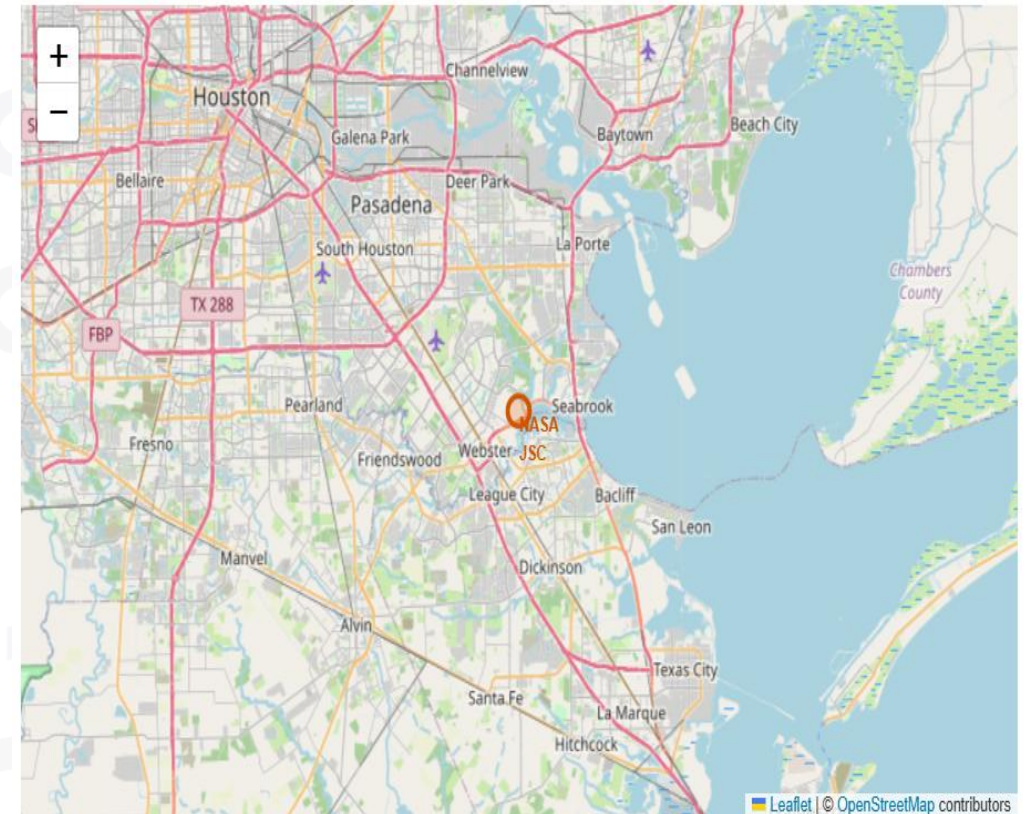


Interactive Map Results - Folium Map

The following slides will show:

- All launch sites marked on a map
- The success/failed launches for each site marked on the map
- The distances between a launch site to its proximities calculated on the map

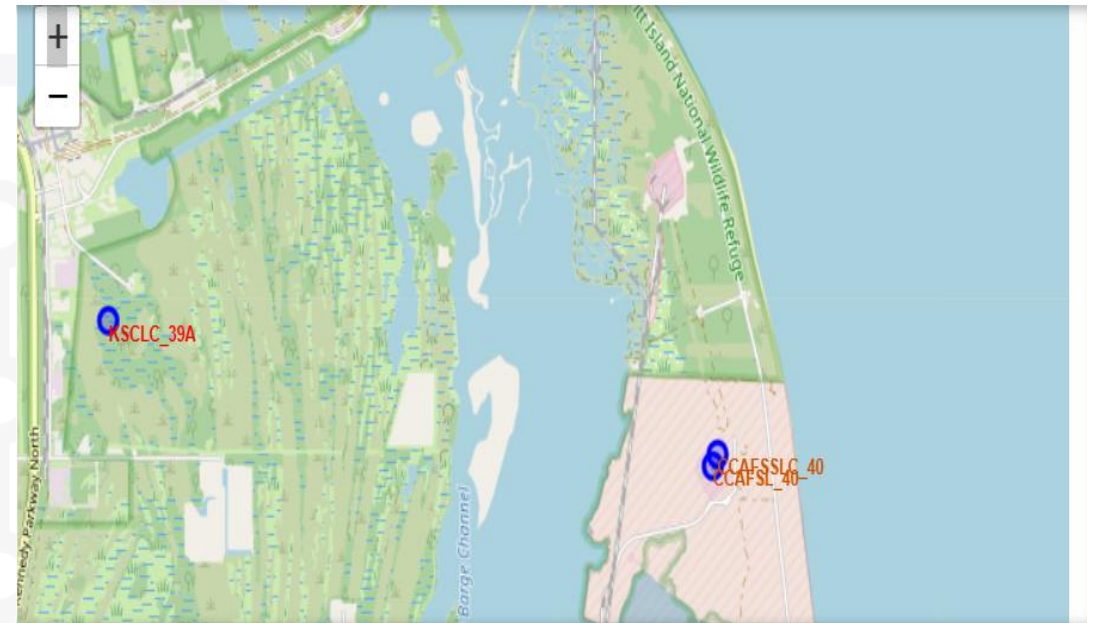
Folium Map object, with an initial center location to be NASA Johnson Space Center at Houston, Texas.



Interactive Map Results - Folium Map

- Folium circle and folium marker for each launch site on the site map

	Launch Site	Lat	Long
0	CCAFS LC-40	28.562302	-80.577356
1	CCAFS SLC-40	28.563197	-80.576820
2	KSC LC-39A	28.573255	-80.646895
3	VAFB SLC-4E	34.632834	-120.610745



Interactive Map Results - Folium Map

- Created markers for all launch records. If a launch was successful (class=1), then we use a green marker and if a launch was failed, we use a red marker (class=0)
- A new column in launch sites dataframe called marker color to store the marker colors based on the class value

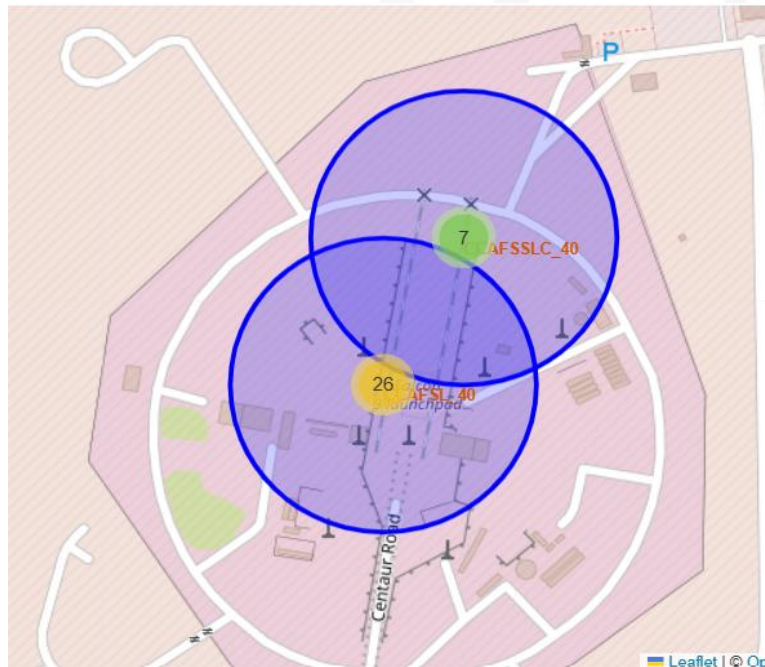
	Launch Site	Lat	Long	class
46	KSC LC-39A	28.573255	-80.646895	1
47	KSC LC-39A	28.573255	-80.646895	1
48	KSC LC-39A	28.573255	-80.646895	1
49	CCAFS SLC-40	28.563197	-80.576820	1
50	CCAFS SLC-40	28.563197	-80.576820	1
51	CCAFS SLC-40	28.563197	-80.576820	0
52	CCAFS SLC-40	28.563197	-80.576820	0
53	CCAFS SLC-40	28.563197	-80.576820	0
54	CCAFS SLC-40	28.563197	-80.576820	1
55	CCAFS SLC-40	28.563197	-80.576820	0

	Launch Site	Lat	Long	class	marker_color
46	KSC LC-39A	28.573255	-80.646895	1	green
47	KSC LC-39A	28.573255	-80.646895	1	green
48	KSC LC-39A	28.573255	-80.646895	1	green
49	CCAFS SLC-40	28.563197	-80.576820	1	green
50	CCAFS SLC-40	28.563197	-80.576820	1	green
51	CCAFS SLC-40	28.563197	-80.576820	0	red
52	CCAFS SLC-40	28.563197	-80.576820	0	red
53	CCAFS SLC-40	28.563197	-80.576820	0	red
54	CCAFS SLC-40	28.563197	-80.576820	1	green
55	CCAFS SLC-40	28.563197	-80.576820	0	red



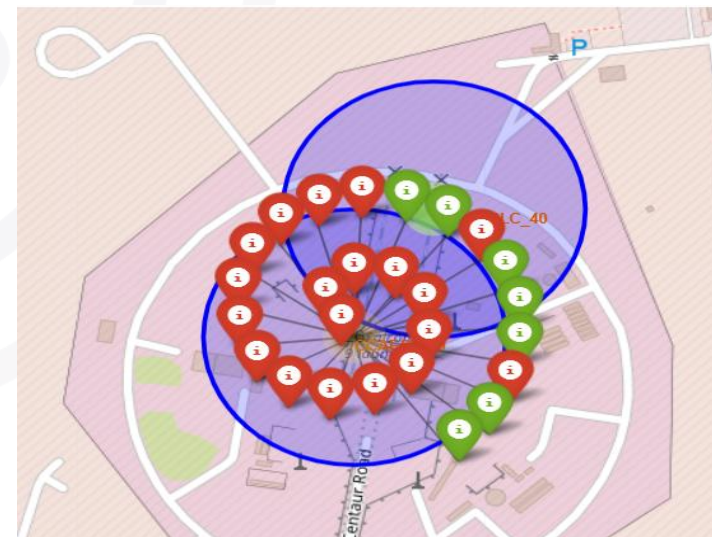
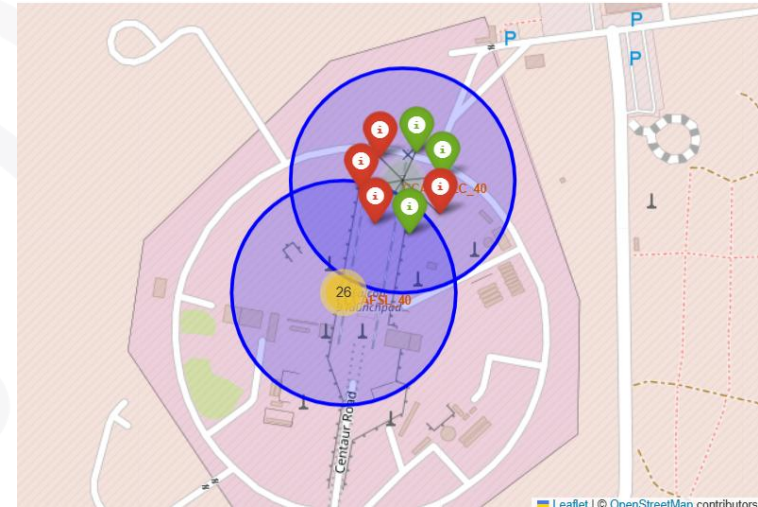
Interactive Map Results - Folium Map

- For each launch result in spacex_df data frame, a folium.Marker to marker_cluster was added.



• 7

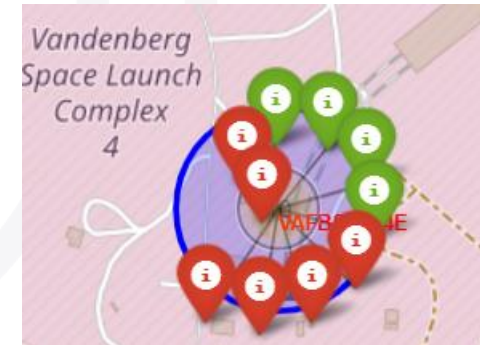
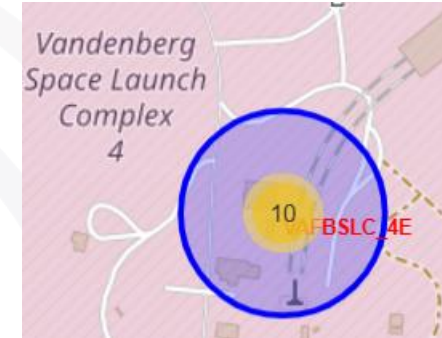
• 26



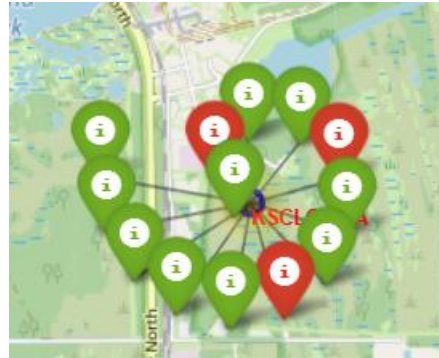
Interactive Map Results - Folium Map

Folium.Marker to marker_cluster was added

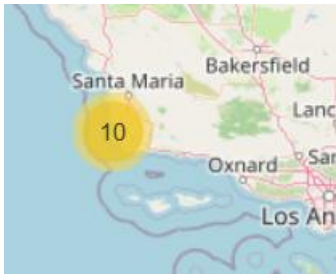
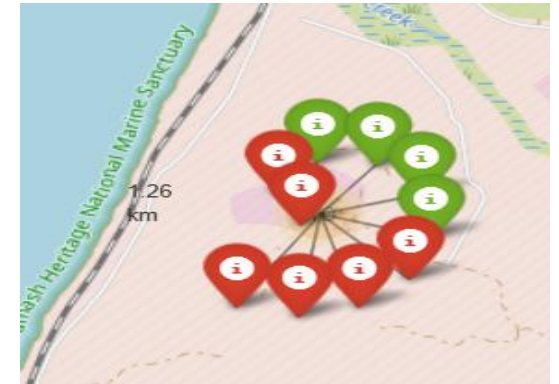
10



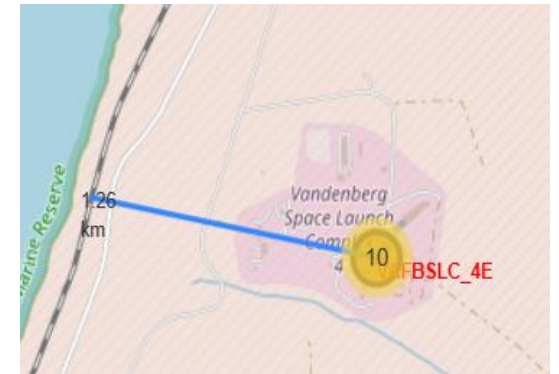
Interactive Map Results - Folium Map



- Calculated distance between a launch site to its proximities

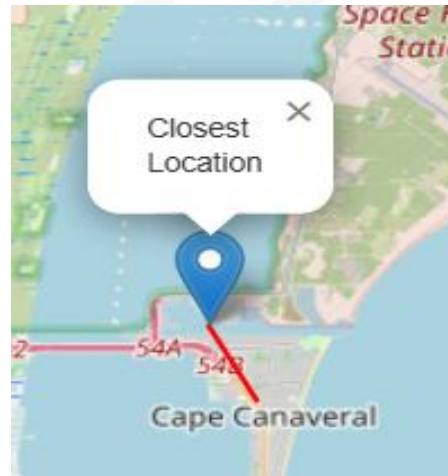


- PolyLine between a launch site to the selected coastline point



Interactive Map Results - Folium Map

Closest Location to KSC LC-39A launch site is Cape Canaveral.



DASHBOARD

The GitHub link - <https://github.com/vkind1/VKEX>

The dash application will contain a dropdown list and a range slider to interact with a pie chart and a scatter plot point chart



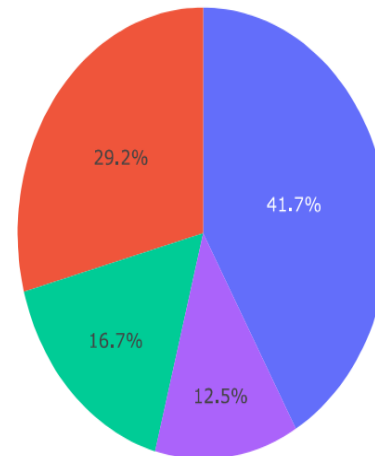
DASHBOARD TAB 1

SpaceX Launch Records Dashboard

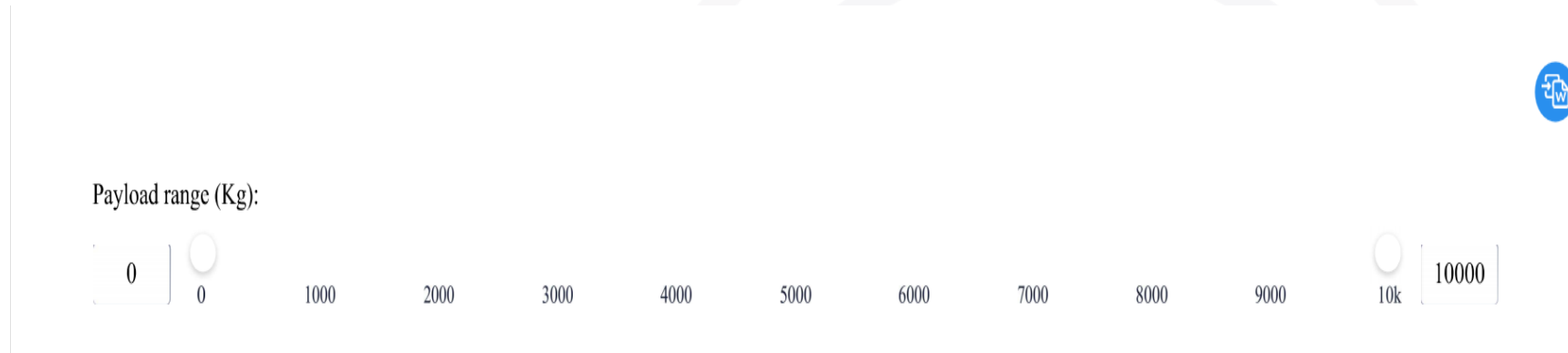
All Sites

× ▾

Share of Successful Launches by Site (%)

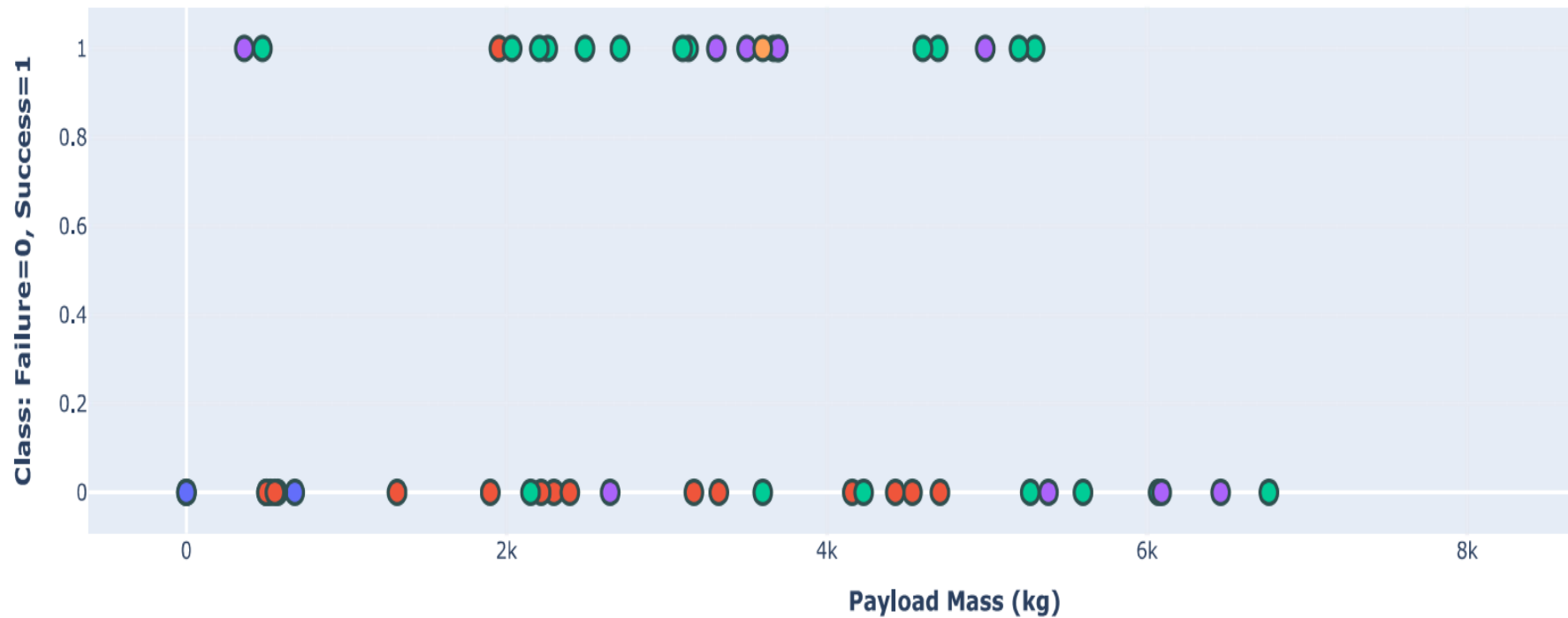


DASHBOARD TAB 2



DASHBOARD TAB 3

Launch outcome v. Payload mass for all sites



DISCUSSION

- Collected Falcon 9 historical launch records from a Wikipedia page titled - List of Falcon 9 and Falcon Heavy launches https://en.wikipedia.org/wiki/List_of_Falcon_9_and_Falcon_Heavy_launches
- Create a dictionary with keys from the extracted column names and converted it into a Pandas dataframe
- In the data set, there are several different cases where the booster did not land successfully. Sometimes a landing was attempted but failed due to an accident; for example,
 - True Ocean means the mission outcome was successfully landed to a specific region of the ocean while
 - False Ocean means the mission outcome was unsuccessfully landed to a specific region of the ocean.
 - True RTLS means the mission outcome was successfully landed to a ground pad
 - False RTLS means the mission outcome was unsuccessfully landed to a ground pad.
 - True ASDS means the mission outcome was successfully landed on a drone ship
 - False ASDS means the mission outcome was unsuccessfully landed on a drone ship.
- The outcomes are converted to Training Labels with
 - 1 means the booster successfully landed
 - 0 means it was unsuccessful.

The data contains several Space X launch facilities: Cape Canaveral Space Launch Complex 40 **VAFB SLC 4E** , Vandenberg Air Force Base Space Launch Complex 4E (**SLC-4E**), Kennedy Space Center Launch Complex 39A **KSC LC 39A** .

- See appendix for chart with the location of each launch and the number of launches for each site.



DISCUSSION

- From cores we would like to learn the outcome of the landing, the type of the landing, number of flights with that core, whether gridfins were used, whether the core is reused, whether legs were used, the landing pad used, the block of the core which is a number used to separate version of cores, the number of times this specific core has been reused, and the serial of the core.
- Requesting rocket launch data from SpaceX API with the following URL: SpaceX
URL="<https://api.spacexdata.com/v4/launches/past>"



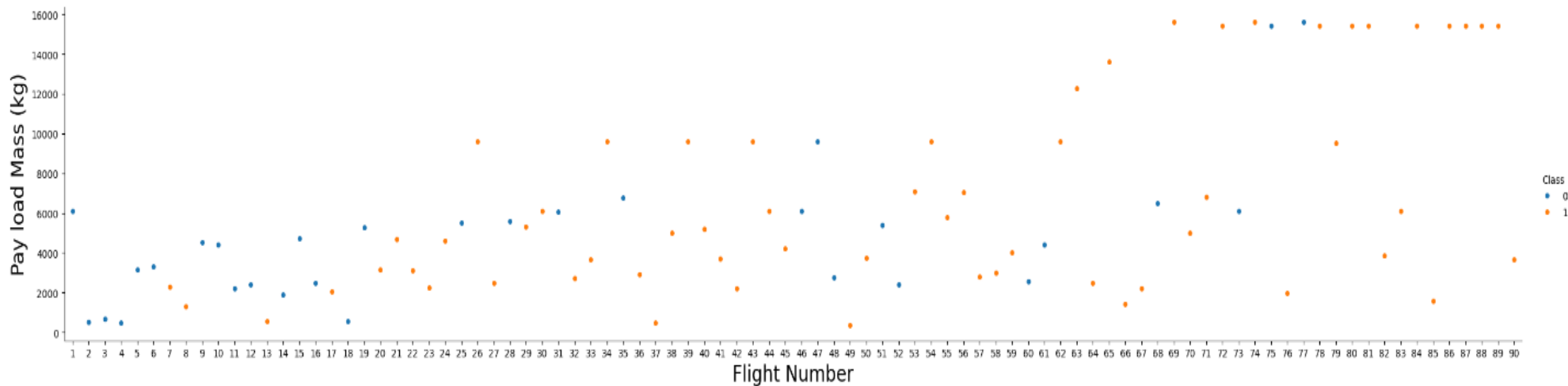
Conclusion

- Companies are making space travel affordable for everyone. Virgin Galactic is providing suborbital spaceflights. Rocket Lab is a small satellite provider. Blue Origin manufactures sub-orbital and orbital reusable rockets. Perhaps the most successful is SpaceX.
- One reason SpaceX can do this is the rocket launches are relatively inexpensive. SpaceX advertises Falcon 9 rocket launches on its website with a cost of 62 million dollars; other providers cost upwards of 165 million dollars each, much of the savings is because SpaceX can reuse the first stage.
- Space Y would like to compete with SpaceX founded by Billionaire industrialist Allon Musk.
- Therefore, if we can determine if the first stage will land, we can determine the cost of a launch.
- The following are some of the questions that were answered and charts with other information are on the following slides.
 - CCAFS 5LC 40 site has had the largest number of successful launches.
 - LEO orbit, success seems to be related to the number of flights
 - The payload range from 2000 to 3500 have the highest launches success rate.
 - The payload range(s) with the lowest launch success rate are 3000 to 7000.



CONCLUSION

EDA with Visualization



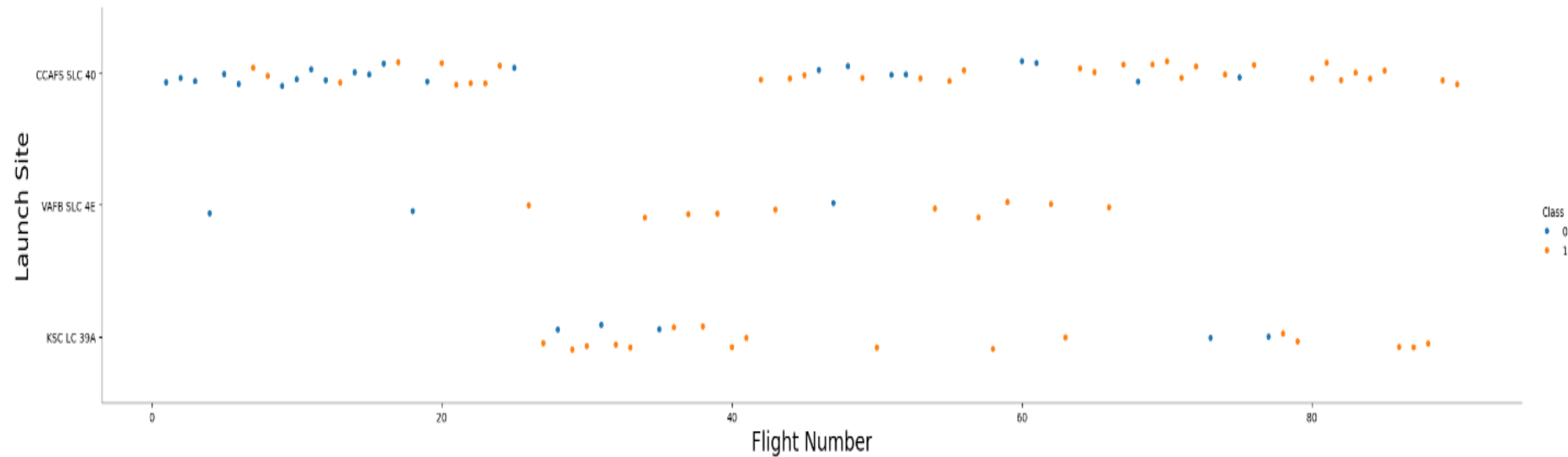
- The chart shows how the FlightNumber (indicating the continuous launch attempts.) and Payload variables would affect the launch outcome.
- When the FlightNumber vs. PayloadMass and overlay the outcome of the launch is plotted out the, you can see that as the flight number increases, the first stage is more likely to land successfully. The payload mass also appears to be a factor; even with more massive payloads, the first stage often returns successfully.



CONCLUSION

EDA with Visualization

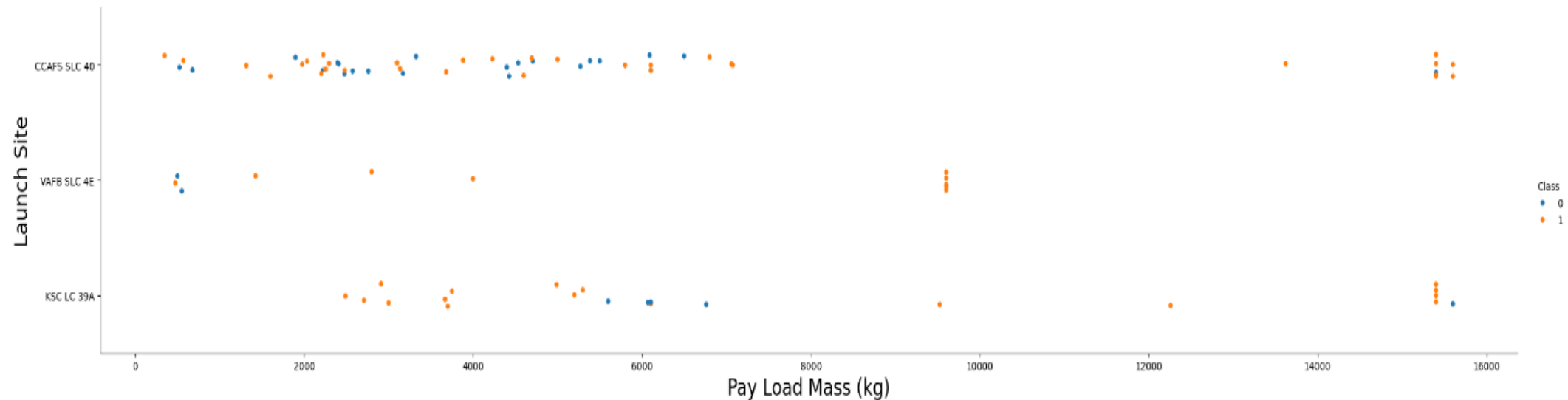
Relationship between Flight Number and Launch Site



CONCLUSION

- EDA with Visualization

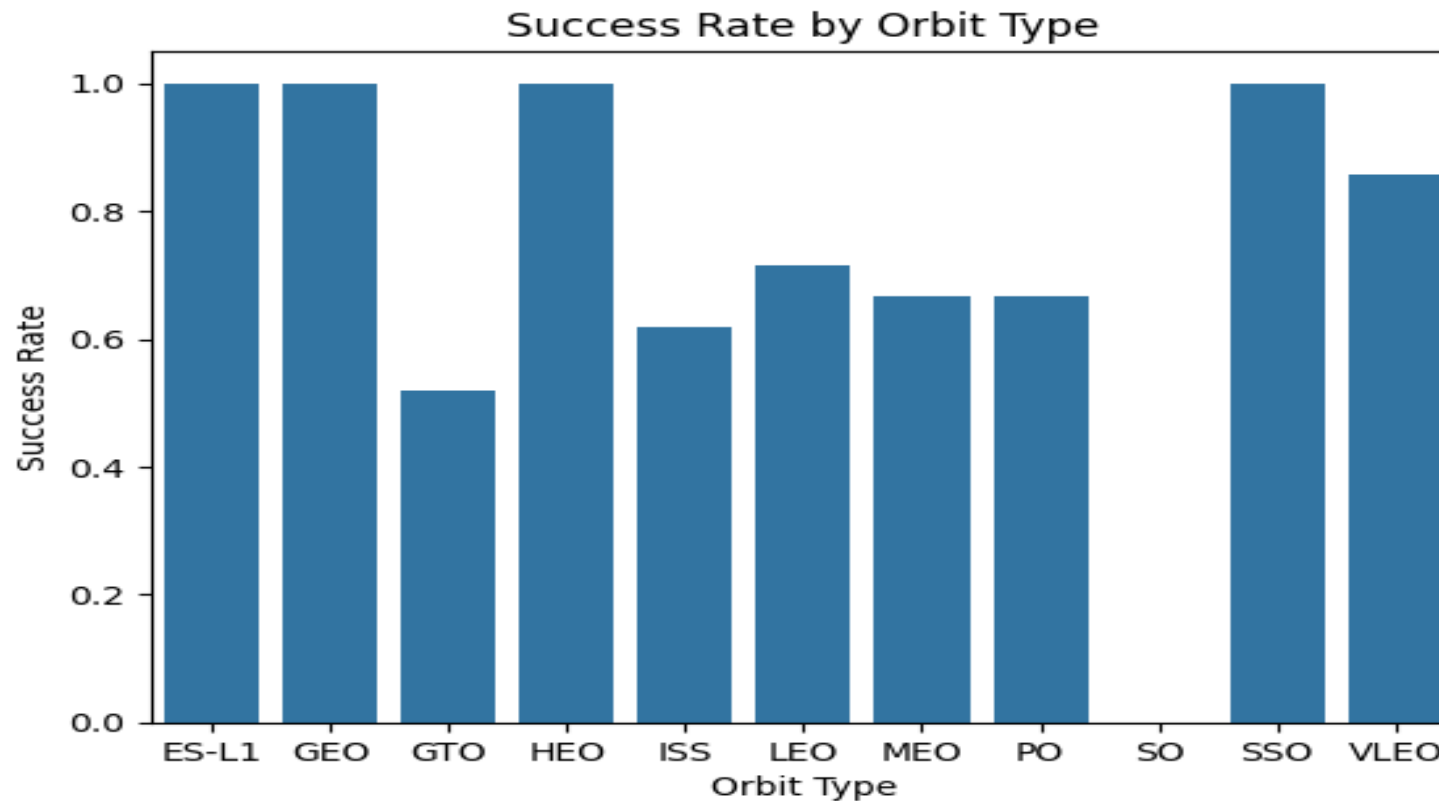
Relationship between launch sites and their payload mass



CONCLUSION

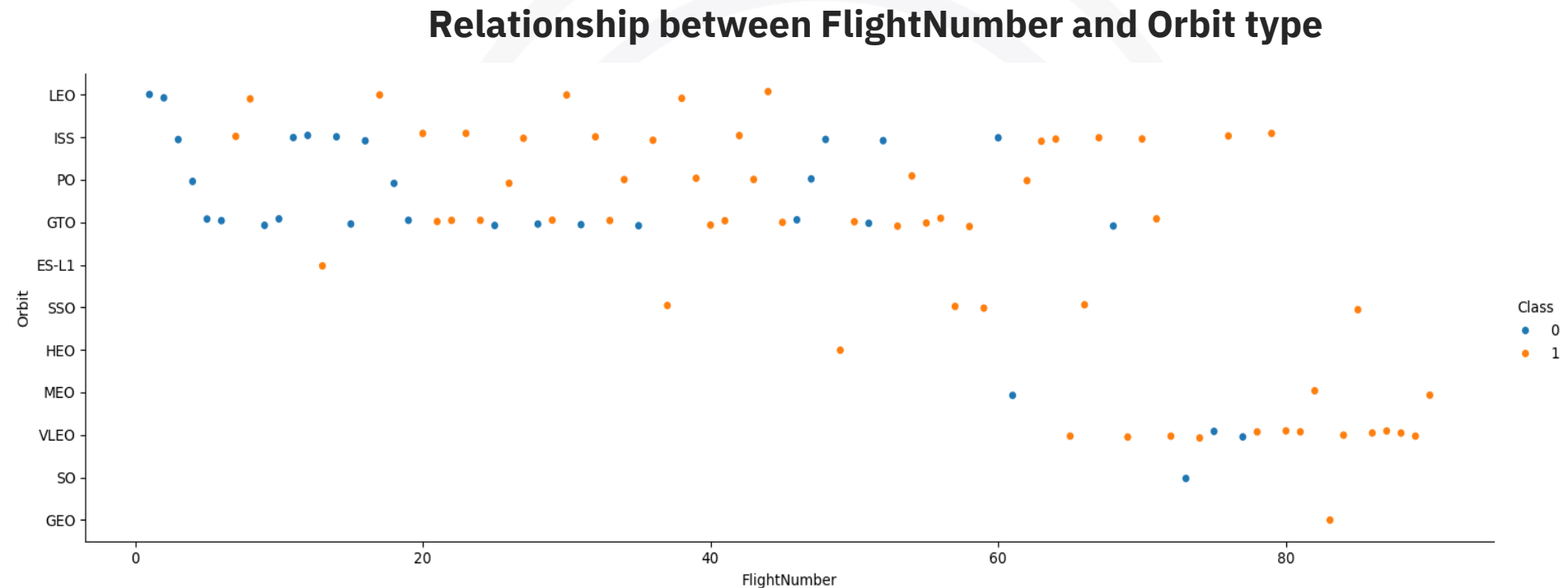
EDA with Visualization

The relationship between success rate and orbit type



CONCLUSION

EDA with Visualization



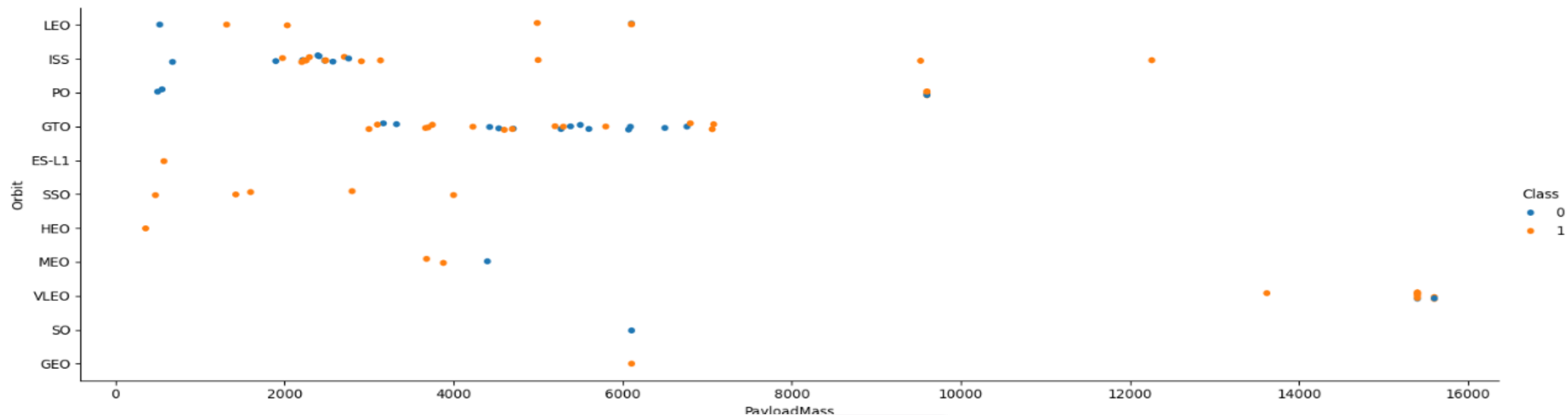
- The LEO orbit, success seems to be related to the number of flights. Conversely, in the GTO orbit, there appears to be no relationship between flight number and success.



CONCLUSION

EDA with Visualization

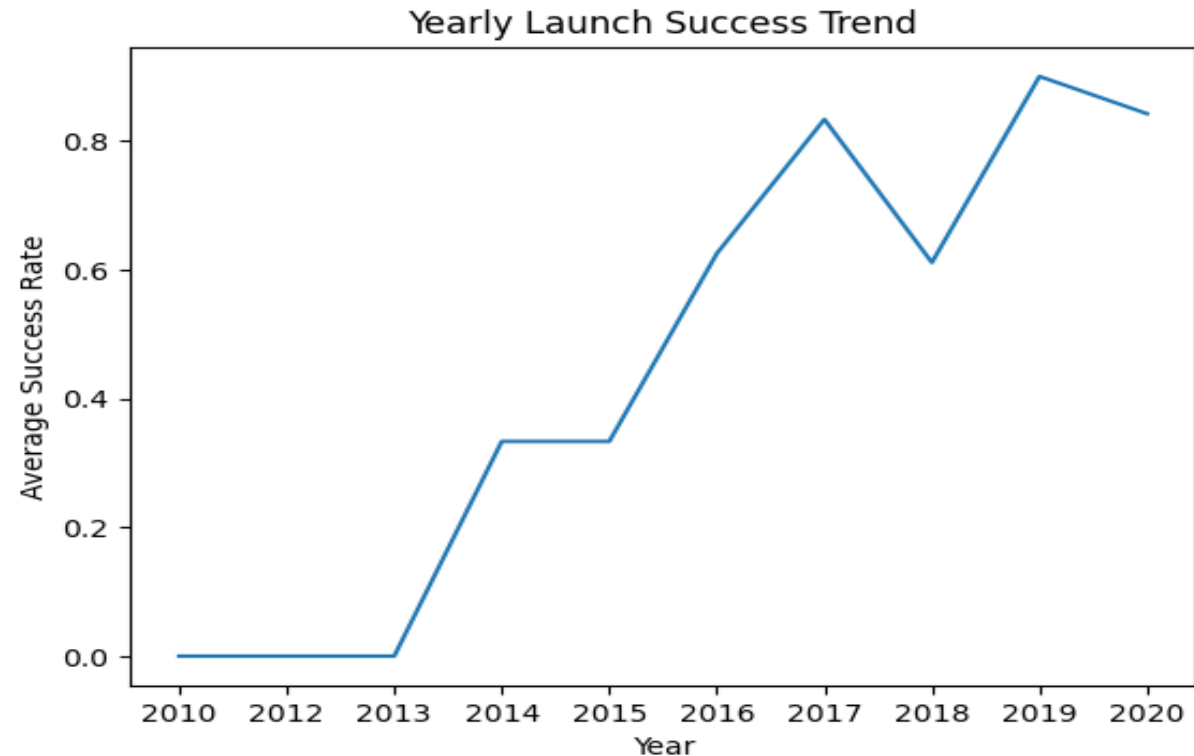
- Relationship between Payload Mass and Orbit type
- With heavy payloads the successful landing or positive landing rate are more for Polar, LEO and ISS.
- However, for GTO, it's difficult to distinguish between successful and unsuccessful landings as both outcomes are present.



CONCLUSION

EDA with Visualization

- The average launch success trend
- The success rate since 2013 kept increasing till 2020



CONCLUSION

EDA with SQL

- Understand the SpaceX DataSet
- Establish a connection with the SQL database and load data into the table
- Connect to the database
- Load the dataset into the corresponding table in a Db2 database
- Execute SQL queries

CONCLUSION

EDA with SQL

```
%%sql
SELECT DISTINCT Launch_Site FROM SPACEXTBL;
```

```
* sqlite:///my_data1.db
Done.
```

Launch_Site

CCAFS LC-40

VAFB SLC-4E

KSC LC-39A

CCAFS SLC-40

Display 5 records where launch sites begin with the string 'CCA'

```
%%sql
SELECT * FROM SPACEXTBL
WHERE Launch_Site LIKE 'CCA%'
LIMIT 5;
```

```
* sqlite:///my_data1.db
Done.
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt



CONCLUSION

EDA with SQL

Display the total payload mass carried by boosters launched by NASA (CRS)

```
%%sql
SELECT Customer, SUM(Payload_Mass_Kg_) AS 'Total PayLoadMass Kg' FROM SPACEXTBL
WHERE Customer = 'NASA (CRS)';
```

```
* sqlite:///my_data1.db
```

Done.

Customer	Total PayLoadMass Kg
NASA (CRS)	45596

Display average payload mass carried by booster version F9 v1.1

```
[14]: %%sql
SELECT Booster_Version, AVG(Payload_Mass_Kg_) AS 'AVG PayLoadMass KG' FROM SPACEXTBL
WHERE Booster_Version = 'F9 v1.1';
```

```
* sqlite:///my\_data1.db
```

Done.

```
[14]:
```

Booster_Version	AVG PayLoadMass KG
F9 v1.1	2928.4

CONCLUSION

EDA with SQL

List the date when the first succesful landing outcome in ground pad was acheived.

Hint: Use min function

```
%%sql
select min(Date), Landing_Outcome from SPACEXTBL
WHERE Landing_Outcome = 'Success (ground pad)'
LIMIT 1;

* sqlite:///my_data1.db
Done.
```

```
: min(Date)    Landing_Outcome
-----
2015-12-22    Success (ground pad)
```

List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000

```
%%sql
SELECT Booster_Version, Payload_Mass_Kg, Landing_Outcome FROM SPACEXTBL
WHERE Landing_Outcome = 'Success (drone ship)'
AND
Payload_Mass_Kg BETWEEN 4000 AND 6000;

* sqlite:///my_data1.db
Done.
```

Booster_Version	PAYLOAD_MASS_KG	Landing_Outcome
F9 FT B1022	4696	Success (drone ship)
F9 FT B1026	4600	Success (drone ship)
F9 FT B1021.2	5300	Success (drone ship)
F9 FT B1031.2	5200	Success (drone ship)



CONCLUSION

EDA with SQL

List the total number of successful and failure mission outcomes

```
%%sql
SELECT
    COUNT(CASE WHEN Mission_Outcome LIKE 'Success%' THEN 1 END) AS 'Successful Missions',
    COUNT(CASE WHEN Mission_Outcome LIKE 'Failure%' THEN 1 END) AS 'Failed Missions'
FROM SPACEXTBL;
```

```
* sqlite:///my_data1.db
Done.
```

Successful Missions	Failed Missions
100	1

List all the booster_versions that have carried the maximum payload mass, using a subquery with a suitable aggregate function.

```
%%sql
SELECT Booster_Version, Payload_Mass_Kg_ FROM SPACEXTBL
WHERE Payload_Mass_Kg_ = (SELECT MAX(Payload_Mass_Kg_) FROM SPACEXTBL);
```

```
* sqlite:///my_data1.db
Done.
```

Booster_Version	PAYLOAD_MASS_KG_
F9 B5 B1048.4	15600
F9 B5 B1049.4	15600
F9 B5 B1051.3	15600
F9 B5 B1056.4	15600
F9 B5 B1048.5	15600
F9 B5 B1051.4	15600
F9 B5 B1049.5	15600
F9 B5 B1060.2	15600
F9 B5 B1058.3	15600
F9 B5 B1051.6	15600
F9 B5 B1060.3	15600
F9 B5 B1049.7	15600



CONCLUSION

EDA with SQL

List the records which will display the month names, failure landing_outcomes in drone ship ,booster versions, launch_site for the months in year 2015.

Note: SQLite does not support monthnames. So you need to use substr(Date, 6,2) as month to get the months and substr(Date,0,5)='2015' for year.

```
%sql
SELECT substr(Date, 6, 2) AS Month, Landing_Outcome, Booster_Version, Launch_Site FROM SPACEXTBL
WHERE Landing_outcome = 'Failure (drone ship)'
AND substr(Date, 0, 5) = '2015';
```

* sqlite:///my_data1.db

Done.

Month	Landing_Outcome	Booster_Version	Launch_Site
-------	-----------------	-----------------	-------------

01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
----	----------------------	---------------	-------------

04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40
----	----------------------	---------------	-------------

Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.

```
%sql
SELECT "Landing_Outcome", COUNT(*) AS Landing_Outcome_Count FROM SPACEXTABLE
WHERE "Date" >= '2010-06-04' AND "Date" <= '2017-03-20'
GROUP BY "Landing_Outcome"
ORDER BY Landing_Outcome_Count DESC;
```

* sqlite:///my_data1.db
Done.

Landing_Outcome	Landing_Outcome_Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1



APPENDIX

Number of Launches from each Site

- CCAFS SLC 40 55
- KSC LC 39A 22
- VAFB SLC 4E 13

•**LEO**: Low Earth orbit (LEO) is an Earth-centered orbit with an altitude of 2,000 km (1,200 mi) or less (approximately one third of the radius of Earth), [1] or with at least 11.25 periods per day (an orbital period of 128 minutes or less) and an eccentricity less than 0.25. [2] Most of the manmade objects in outer space are in LEO [1].

•**VLEO**: Very Low Earth Orbits (VLEO) can be defined as the orbits with a mean altitude below 450 km. Operating in these orbits can provide a number of benefits to Earth observation spacecraft as the spacecraft operates closer to the observation [2].

•**GTO** (Geostationary Transfer Orbit): A geostationary transfer orbit is an elliptical Earth orbit used to transfer satellites from low Earth orbit (LEO) to geostationary orbit (GEO). In a GTO, the perigee (closest point to Earth) is much lower than GEO altitude, while the apogee (farthest point) reaches approximately 22,236 miles (35,786 kilometers) above Earth's equator — the altitude of a geostationary orbit. Satellites in GTO use onboard propulsion to circularize their orbit at GEO altitude, where they can provide services such as weather monitoring, communications, and surveillance. [3].

•**SSO (or SO)**: It is a Sun-synchronous orbit also called a heliosynchronous orbit is a nearly polar orbit around a planet, in which the satellite passes over any given point of the planet's surface at the same local mean solar time [4].

•**ES-L1**: At the Lagrange points the gravitational forces of the two large bodies cancel out in such a way that a small object placed in orbit there is in equilibrium relative to the center of mass of the large bodies. L1 is one such point between the sun and the earth [5].

•**HEO** A highly elliptical orbit, is an elliptic orbit with high eccentricity, usually referring to one around Earth [6].

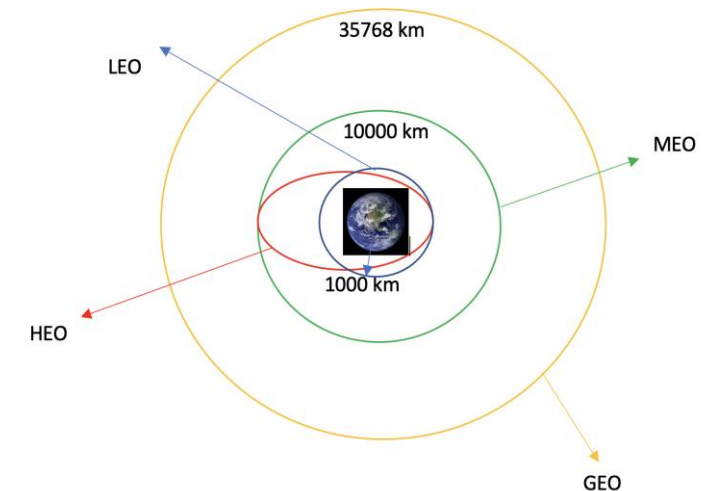
•**ISS** A modular space station (habitable artificial satellite) in low Earth orbit. It is a multinational collaborative project between five participating space agencies: NASA (United States), Roscosmos (Russia), JAXA (Japan), ESA (Europe), and C (Canada) [7].

•**MEO** Geocentric orbits ranging in altitude from 2,000 km (1,200 mi) to just below geosynchronous orbit at 35,786 kilometers (22,236 mi). Also known as an intermediate circular orbit. These are "most commonly at 20,200 kilometers (12,600 mi), or 20,650 kilometers (12,830 mi), with an orbital period of 12 hours [8].

•**HEO** Geocentric orbits above the altitude of geosynchronous orbit (35,786 km or 22,236 mi) [9].

•**GEO** It is a circular geosynchronous orbit 35,786 kilometres (22,236 miles) above Earth's equator and following the direction of Earth's rotation [10].

•**PO** It is one type of satellites in which a satellite passes above or nearly above both poles of the body being orbited (usually a planet such as the Earth [11] some are shown in the following plot:



APPENDIX

True Ocean means the mission outcome was successfully landed to a specific region of the ocean while False Ocean means the mission outcome was unsuccessfully landed to a specific region of the ocean. True RTLS means the mission outcome was successfully landed to a ground pad False RTLS means the mission outcome was unsuccessfully landed to a ground pad. True ASDS means the mission outcome was successfully landed to a drone ship False ASDS means the mission outcome was unsuccessfully landed to a drone ship. None ASDS and None None these represent a failure to land.

create a set of outcomes where the second stage did not land successfully:

- The number and occurrence of each orbit

Orbit

GTO	27
ISS	21
VLEO	14
PO	9
LEO	7
SSO	5

Orbit

Leo	7
SSO	5
MEO	3
ES-L	1
HEO	1
SO	1
GEO	1



APPENDIX

Using the Outcome, a list was created where the element is zero if the corresponding row in Outcome is in the set bad outcome; otherwise, it's one. Then assign it to the variable landing class

If the value is zero, the first stage did not land successfully; one means the first stage landed Successfully

the success rate: `np.float64(0.6666666666666666)`

class

0	0
1	0
2	0
3	0
4	0
5	0
6	1
7	1

- The launch success rate may depend on many factors such as payload mass, orbit type, and so on. It may also depend on the location and proximities of a launch site, i.e., the initial position of rocket trajectories. Finding an optimal location for building a launch site certainly involves many factors and hopefully we could discover some of the factors by analyzing the existing launch site locations.



APPENDIX

- The coordinates for each site.

Launch Site	Lat	Long
0	CCAFS LC-40	28.562302 -80.577356
1	CCAFS SLC-40	28.563197 -80.576820
2	KSC LC-39A	28.573255 -80.646895
3	VAFB SLC-4E	34.632834 -120.610745